

SPECTRA VIDEO

USERS GROUP -- Wgtn

Volume 4, Number 1 -- July/August 87

The next meeting is on Monday, August 17, 1987 at 7.30pm in the Staff room of Newlands Intermediate, Bracken Rd, Newlands. The September meeting will be the 21 September, 1987. (ie as usual the 3rd Monday of the month.)

Wellington SV Users Group

Committee Members

Chairman	-----	Jim Oliver	699-618
Secretary/Treasurer	-----	Ross Fletcher	788-454
Librarian	-----	Scotty Baxter	784-619

Newsletter Committee

Steve Hanson	843-495
Joseph Dol	889-275
Phillip Daymond	

Other areas of responsibility for committee members are:

- Program evaluation
- Jim -- Public Domain BASIC & Radio Access
- Ross -- Public Domain CP/M

Club Subscriptions

Just a reminder to check the details on your newsletter label.

Please let us know if any of the details are incorrect, and please note the date in the bottom left-hand corner ...

This is the EXPIRY DATE of your subscription

Club Subscriptions

It is your subscriptions which keep the club running, therefore if you want the support of the Group to continue, please help us by checking your subscription renewal date.

If your subs are due before October, please forward them to the Club's P.O. Box 26050 - Newlands, WELLINGTON.

Remember - the subs are still only:

\$18.00 For NZ Residents

NZ\$30 For Australasian Subscriptions

NZ\$40 For other areas overseas.

PUBLIC DOMAIN BASIC - AND RADIO BROADCAST

We have been having some trouble recently getting the monthly Radio Access Broadcasts back 'on the Air'. But with perseverance we got back to our normal 'First Saturday of the Month' Slot for the 2 August. For future broadcasts, keep an eye on the Listener which should indicate when our broadcast of Public Domain Software will go to air, but we hope to keep to the first Saturday of each month - so please bear with us.

- - RADIO ACCESS - 2YB - 789kHz

If you caught the Broadcast on 1 August, you will have a copy of the Public Domain Spreadsheet PD-CALC.

The documentation on following pages goes with this program ..

(Note: if you didn't or can't receive the Public Domain Software broadcast, you can write to Jim through the Club P.O. Box for a copy of the BASIC program)

PUBLIC DOMAIN SOFTWARE

Four new tapes have now been added to the list of P.D Software available. The contents of these tapes are listed below. The previous tape descriptions have been published in the December 1985 and September 1986 issues. The disc based software follows these lists fairly closely and will give you a good idea of what is available.

Note: Programmes from this list are broadcast from Access radio on the first Saturday of every month on station 2YB 783kHz.

=====>

TAPE 15 This tape is a real mixed bag and contains 21 programs. Input prog.from Aust.Users group, a sound tutour, morgage calculations, an excellent disc scanner/doctor, an adventure game, 3 graphics progs., worm growing game, 2 letter puzzles, Yahtze, 2 action games, musical tunes, A.I type prog., vram demo, error codes, interest calc., print screen prog.

TAPE 16 A new set of 13 excellent games, some from the Austrian Users Gr. Joust game, a fast tiler game, labyrinth, sound effects library utility, golden flute game, hexapawn, and others including a Donkey Kong!

TAPE 17 This one follows on from No. 16 and presents 8 new games and 2 useful utilities. Some are in German. An A.I (learning) Hexapawn game which allows variable board sizes. A fast Q-Bert game, zap a gopher, stocks & shares, Adventure game, function plotter and a disc file utility.

TAPE 18 11 miscellaneous progs. A learners chess game, vram demo, Basic version of CP/M Nsweep, graphics pad prog, gunnery game, caverns of Venus game, rooms maze adventure, bombing game, compatability prog., baccarat card game.

We have now converted about 17 MSX games and utilities over to SVI. These programmes include the following:

Compatability, library file utility, bombing game, baccarat and black jack card games, a moon lander, shootout at OK corral, Simon game, cash account utility, helicopter game, bowling game, 2 more games and an adult shooting game.

At this stage the MSX programmes are available on disc only except for the last 3 on tape 18. Some of these programmes require some additional debugging but all will run. Some are very good especially the graphics type games.

Software can be ordered from : SV PD Software
P.O. Box 26050
Newlands, Wellington,
New Zealand

The costs are as follows: tapes and discs are \$3 each with a handling charge of \$2.50 per order. Please make cheques payable to J.Oliver.

We have found that some disc drives are very 'format sensitive' and to avoid problems you must supply discs formatted on your own drives. You may get extra bonus progs. on disc as well as those indicated above due to the extra space available.

PD-CALC

DESCRIPTION:

PD-CALC is a user friendly calculator type spreadsheet. It runs on a SPECTRAVIDEO 328 (or 318 with 16K or 64K ram cartridge installed) in 40 column mode. Data can be sent to a printer if desired and can be saved on disk or cassette.

PD-CALC was designed primarily for those users who do not have 80 column capability but wish to run a spreadsheet type program. While it may not be as sophisticated as systems available to CP/M users, PD-CALC is never the less quite powerful and most importantly, very easy to use.

Numerical data is accepted via the keyboard in INPUT mode and several types of calculation can be carried out. Multiplication, division, addition, subtraction, percentage and average routines can be accessed in COMPUTATION mode. Other modes are; DELETE, SORT, SUMMATION, PRINT, CLEAR MEMORY, LOAD and SAVE.

While PD-CALC operates in 40 column mode on the UDU, two screens are available which in effect give you 80 columns. (See Figs. 2 & 3). Data can be written to and exchanged between screens. When a hardcopy is required, the printer prints the data out in 80 column format. (See Fig. 1).

As previously mentioned, PD-CALC will run on either cassette or disk based systems but is written initially to save and load to disk. Minor changes to the program are required if you wish to run it on a cassette based system. They are as follows:

```
Change line 1690 to read;
1690 LOCATE0,2;PRINT"INSERT DATA CASSETTE IN RECORDER"
Delete line 1697
Change line 1710 to read;
1710 OPEN"CALC"+FL$FORINPUT AS#1
```

```
Change line 1845 to read;
1845 LOCATE0,2;PRINT"PLACE EMPTY CASSETTE IN RECORDER"
Change line 1870 to read;
1870 OPEN"CALC"+FL$FOROUTPUT AS#1
```

XXN.B. Lines 2570 and 2650 print down arrows and up arrows to printer. Also, printer command codes within lines 2510 to 2660 support an NEC PC-8023B-C dot matrix printer. Changes may need to be made here to suit your printer. They are as follows;

LPRINTCHR\$(27);CHR\$(69) in line 2540 produces Elite printing mode. (12cpi). LPRINTCHR\$(14) and LPRINTCHR\$(15) in line 2545 produce enlarged (5cpi) print and back to pica (10cpi) respectively. LPRINTCHR\$(10) in line 2640 returns the printhead to the left margin with a line feed.

INSTRUCTIONS:

If you intend using PD-CALC with a disk based system, simply copy directly to disk. If however you wish to use cassettes, first make changes to the program as previously outlined.

LOAD and RUN PD-CALC. You will be presented with Screen 1. (See Fig.2). The marker block is in position A1. Move the block around using the cursor keys. At this stage, PD-CALC is in INPUT mode. To access Screen 2, press function Key F10. (See Fig.3). To return to Screen 1, press F9.

INPUT MODE:

Position the block in a suitable position, e.g. A1. Enter a number. Move the block to A2 and enter another number. Continue down column A until you have say, 12 numbers.

DELETE MODE:

Make sure that the block is within the column you are working in, in this case, column A. Now press F1. You are now in DELETE Mode. At the bottom of the screen you will see L1? and below that L2? L1 is the location you wish the deletion to commence from and L2 the end location.

Enter 4 in answer to L1? and press <ENTER>. Enter 11 in answer to L2? (You will not need to use <ENTER> this time because the location consists of more than one digit.) Locations A4 to A11 are now clear. To delete a single number, either enter the same location in answer to L1 and L2 or alternatively, place the block over the desired number and press the <SPACE BAR>.

SUMMATION:

Enter some more numbers in Column A. Then press F2 to get into SUMMATION mode. (Make sure that the block is still within the column you are working with.) Enter locations for the start and end of the column you wish to sum over. Now move the block to wherever you wish the result to be placed and press <SPACE BAR>. (Do not hold the Space Bar down too long or the answer will be deleted.)

SORT MODE:

Follow the same procedure as in SUMMATION, excluding the Block move. The column of figures will be sorted with low numbers at the top down to high numbers at the bottom.

COMPUTATION MODE:

COMPUTATION only works on two locations within the same column at any one time. Functions are + - / * % and A (Average). The functions + - and * are straight forward. Enter the cell

locations in either order. When using functions / (Division) and %, remember that location 1 is the location of the number that you wish to be divided and in the case of % (Percent), the number the percentage is to be taken from. Location 2 of course is for the number you are either dividing by or working your percentage from. When using the AVERAGE function, enter the start and end locations of the figures to be averaged over, then move the block to a desired cell and press the (SPACE BAR). (I hope that you aren't too confused yet!)

Try filling up a few columns on both screens and also placing results from one screen to the other.

SAVE MODE:

Now that you have become familiar with the operation of PD-CALC, you may wish to SAVE the contents to either Disk or Cassette. Simply press F6 and follow the prompts.

CLEAR MEMORY:

Having saved your work, press F7 to clear the contents of PD-CALC from memory. Again, just follow the prompts.

LOAD MODE:

Insert your Data Disk or Cassette and press F5. Your data will be loaded back into memory and presented on screen.

PRINT MODE:

If you have a printer, set it up and press F8. A full printout of the contents of PD-CALC will be produced.

That's really all there is to PD-CALC. As I mentioned before, it is very simple to use! Have Fun!!

	A	B	C	D
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				
21				
22				
23				
24				
25				
26				
27				
28				
29				
30				
31				
32				
33				
34				
35				
36				
37				
38				
39				
40				
41				
42				
43				
44				
45				
46				
47				
48				
49				
50				
51				
52				
53				
54				
55				
56				
57				
58				
59				
60				
61				
62				
63				
64				
65				
66				
67				
68				
69				
70				
71				
72				
73				
74				
75				
76				
77				
78				
79				
80				
81				
82				
83				
84				
85				
86				
87				
88				
89				
90				
91				
92				
93				
94				
95				
96				
97				
98				
99				
100				

Row: 1
Mode: INPUT

F1=Del. F2=Sum.
F3=Com. F4=Int.
F5=Loc. F6=Save
F7=Clear F8=Print
F9=Set. F10=Set.

Fig.2 Screen 1

<<<<< PDCALC >>>>>

	A	B	C	D	E	F	G	H
1:	1456.00	3456.00	4567.00	1234.00	2341.00	6543.00	8765.00	9345.00
2:	57.00	78.00		8521.00	875.00			
3:	8.34	542.00		52.00	589.00			
4:	28.00	52.00	549.00	98.00	542.00			
5:	34.00			84.00		5.00		85.00
6:	871.98			84.00		21.56		548.00
7:	45.00		34.00	71.00		48.00		587.00
8:	87.35		68.00	45.00		67.00		25.00
9:	46.78		96.00			87.98		54.00
10:	79.00		34.00			563.00		258.00
11:	7854.26		879.00		792.54			4587.00
12:	569.98		0.45					5.00
13:	258.00		87.00					0.23
14:	78.00		7992.00					0.26
15:	85.00		67.12				58.00	0.25
16:	785.00						54.00	7.26
17:	963.00						876.00	84.00
18:	89.00							128.36

Fig.1: Sample printout of PD-CALC

Row: 1

Mode: INPUT

F1=Del. F2=SU
F3=Load F4=PSU
F7=Clear F8=PAUSE
F9=SI. F10=SE.

Fig.3 Screen 2

Public Domain CP/M

Anyone wanting anything from the library of CP/M PD software, please write to Ross Fletcher at the club address, specifying either which disc, or which programs you would like. The listings in previous issues are printed from double sided disks therefore as most drives are single sided, a whole disc from the list now occupies up to 2 full discs. Most of the programs on the discs are 'Squeezed', this means they have been run through a function in New Sweep (NSW, Nsweep etc. but not the earlier Sweep) so as to take up as little room as possible. They all have a 'Q' as the 2nd letter in their extension, and can be easily unsqueezed by using New Sweep to tag them and then the 'Q' option to unsqueeze them, usually onto another disc. Please send discs, rather than asking us to purchase them for you, and please send them formatted. Please also include, either a stamped return mailer, or add 60c to your payment to cover postage. We will return your disk in your packaging.

Newsletter Contributions

HERE IS
YOUR OPPORTUNITY
TO GET INTO ----> PRINT

There are a lot of you who are writing programs in BASIC that other people would be interested in, so **PLEASE** send in a brief outline of your program together with your program (on cassette or disk and we can merge it into the newsletter) care of the club PO Box. Any of the following items would be gladly received:

- Programs - Basic, Pascal, Assembly, You Name It.
- Programming hints, and techniques.
- Utilities - any type, as long as they are self explanatory or come with instructions.
- News - anything you think other club members might like to know.
- Problems - we will find someone to answer them if we can't.

Everything we receive will be considered. If you are sending in printed copy (we prefer a disk or tape file) make sure that

1. It is printed on A4 paper with 6 lines at the top and bottom and 1 inch on either side.
2. you use a ribbon that gives dark print or use emphasised mode

otherwise it will not be possible to print your article.

QUESTIONS AND ANSWERS

Our May/June issue posed a couple of questions from a dBASE II user, for which John Hodgson has supplied answers:

1. How to use the 'RESET B' command to swap the disks in drive B. (Apparently the dreaded 'Bdos Err on B; R/O' message was the result)

A. John points us to the dBASE Manual which says "The RESET command is used to reset the CP/M bit map after a diskette has been swapped. Normally ... (requires soft boot) ... (etc.). RESET attempts to re-open all files which were open prior to the swap. If a file that was open is no longer mounted on an active disk drivem RESET closes the file internally.
WARNING If a disk is swapped that contains a file with the same name as a file that was previously open, the RESET operation will erroneously NOT close that file. This condition can be avoided by closing all non-essential files prior to the swap and subsequent reset command. A USE command with no filename will close the file in use, a CANCEL command will close any command files that may be open.
Issuing a RESET command when no disk swap has taken place has no effect."

NOTE: A version of dBASE II currently floating around , Version 2.3C, was withdrawn by ASHTON TATE some weeks after its release. The bugs etc were addressed in the enhanced Version 4.xx series. I am using Version 4.1 and the RESET command works except that dBASE prompts come up as garbage - something obviously goes wrong with the monitor settings, ... maybe! It actually ran my program except I wasn't game to allow a file update. Also I operate from drive B: which may or may not have a bearing ... Best of Luck!

2. When our local member tries to quit from dBASE II using the .. QUIT TO <com file list> option (ie QUIT TO WS) .. A '*** SYNTAX ERROR ***' is returned.

A. John tells me that this introduces two features of dBASE.. First the correct format for a QUIT TO command is...
QUIT TO "FILENAME" - (note the quotation marks)
Secondly, there is an extension which allows you to automatically return to dBASE after running the <com file>
QUIT TO "WS1", "DBASE" - (ie run "WS1" then "DBASE")
or to run a particular <.CMD> file when re-returning to dBASE ...

QUIT TO "WS1", "DBASE START" - (ie run "WS1" then 'START' cmd under dBASE)

It would appear that the QUIT TO .. command sets up a type of SUBMIT file which could make for limitless options.

STOP PRESS

WE HAVE NOW CLOSED OUR COMPETITION -- -- --

'THE GREAT 'SWEEP' COMPETITION' TO BE PRECISE.
THE WINNER - WHO WILL RECEIVE A BOX OF DISKS FOR HIS TROUBLE -
WAS BRYAN FISK.

A REVIEW OF THE PROGRAM IS IN THE PIPELINE BUT, REGRETTABLY
DID NOT MAKE THE BULLETIN DEADLINE - YOU WILL SEE IT NEXT
TIME. BRYAN'S PROGRAM SHOWED IMAGINATIVE USE OF BASIC TO
ACHIEVE A VERY GOOD FILE HANDLING UTILITY FOR A SINGLE DISK-
DRIVE SYSTEM. THE PROGRAM IS NOW IN THE PUBLIC DOMAIN, AND
YOU SHOULD GET A COPY.

==) ANY MORE IDEAS FOR ANOTHER COMPETITION - LET US KNOW (===

SOME MORE QUESTIONS FOR THE THRONG:

1. A NEW SV OWNER IS HAVING TROUBLE 'ESCAPING' FROM GAMES
PROGRAMMES SUCH AS ARMOURD ASSAULT ..
FOR A SHORT ANSWER - PERHAPS SOMEONE WOULD LIKE TO EXPLAIN THE
PROGRAM PROTECTION SYSTEM WHICH PREVENTS A SIMPLE USE OF A
CONTROL STOP (OR ESCAPE KEY AS WAS TRIED BY THEM).

2. A MEMBER HAS RUN INTO THE DREADED 'BAD ALLOCATION TABLE'
ERROR WITH A VERY LARGE FILE ON A DOUBLE-SIDED BASIC DISK.
HE WOULD LIKE SOMEONE TO EXPLAIN HOW TO RECALL EACH TRACK BACK
ONTO THE SCREEN TO SAVE IT ONTO ANOTHER DISK - OR DOES SOMEONE
KNOW OF A PROGRAM (OR WANT TO WRITE ONE) WHICH CAN DO THIS
RECOVERY OF A PROGRAM - WITH ALL THE EXPLANATIONS ON HOW TO
PUT IT INTO OPERATION?

THE NEXT QUESTIONS COME FROM ANOTHER RELATIVELY NEW MEMBER:

3. PLEASE CLARIFY THE PROCEDURE FOR READING A SEQUENTIAL FILE
(CONSISTING OF A NUMBER OF RECORDS OF STRING VARIABLES ALL OF
FIXED LENGTH), AND USING THE (EOF) FUNCTION, SO THAT THE YOU
DO NOT GET THE 'INPUT PAST END OF FILE' ERROR.
THE MEMBER SOLVED THE PROBLEM TEMPORARILY BY READING TWO
ADDITIONAL SPACES AT THE END OF EACH RECORD (IE THE END OF
EACH LOOP IN THE PROGRAM) USING B\$=INPUT\$(2,1). IS THE
PROBLEM WHICH SHOWS THE ABOVE ERROR MESSAGE CONNECTED WITH
FILE CONTROL CHARACTERS? (THE PROBLEM ISN'T APPARENT IN
NUMERIC FILES.

4. WITH RANDOM FILES, CAN YOU USE THE 'EOF' FUNCTION. IF SO,
HOW? - (THIS HAS BEEN TRIED BUT GIVES THE 'INPUT PAST END'
MESSAGE.

5. IS THERE ANY OTHER WAY TO ACCESS A RANDOM FILE OTHER THAN
USING THE 'GET (RECORD NO.)' STATEMENT, IE USING A NAME OR
OTHER REFERENCE?

6. IS IT POSSIBLE TO WRITE A RECORD LONGER THAN 128 BYTES TO
A RANDOM DISK FILE?

SUBROUTINE OVERLAYS

Bob Curwood Christchurch

```

1 GOTO 10
2 A=&H8001
3 PRINT"Line#";VAL("&H"+HEX$(PEEK(A+3))+STRING$(2-LEN(HEX$(PEEK(A+2))),4&)+HEX$(
PEEK(A+2)))TAB(12)" is at &H"+HEX$(A):A=VAL("&H"+HEX$(PEEK(A+1))+STRING$(2-LEN(HEX$
$(PEEK(A))),4&)+HEX$(PEEK(A)))":IFPEEK(A)=0THENPRINT"Program ends at &H"+HEX$(A):F
NDELSE3"Curwood
10 PRINT"This is the main program"
15 INV$=CHR$(27)+"p":NOMS=CHR$(27)+"q"
20 PRINT"It can be any length"
30 PRINTINV$"The subroutine at line 1000 is the one that will be overlayed by th
e routines called "CHR$(34)":sub1"CHR$(34)" (or 2/3/4 etc)"NOMS:FORA=1TO1000:NE
VT
40 GOSUB1000
50 BLOAD":sub1"
60 GOSUB1000
70 PRINTINV$"This is the main program"NOMS
80 BLOAD":sub2"
90 GOSUB1000
100 PRINTINV$"This is the main program"NOMS
110 BLOAD":sub3"
120 GOSUB1000
130 PRINTINV$"This is the main program"NOMS
150 BLOAD":sub4"
160 GOSUB1000
170 PRINTINV$"This is the main program"NOMS
180 END
1000 CLS:PRINT"This is the original subroutine and must be a long routine (or
padded out with REM lines full of something) to make sure there is enough ro
om at the end of the program for the longest
1010 PRINT"subroutine to go in the same space in memory without overlaying the
variable table.
1020 PRINT"Remember that the bload used to get the new subroutines into memory ca
lls in 1 or more multiples of 255 bytes so if the longest subroutine is (say)
3.5 pages long (a page is 255 bytes) then THIS subroutine must be at least 4
pages long
1030 PRINT"Press any key":AS=INPUT$(1):CLS
1040 PRINT"This would be a good place to put the instructions for the program a
s they would (presumably) only be needed when the program is first run
1050 PRINT"This subroutine, as written, is 8.458 pages long, I.E 2157 bytes
1060 REM This example has the subroutines loading at &h82DA and they can be up t
o 8 pages long. By using the linepeek routine with GOTO2 you can see where everyt
hing is.
1070 REM For this example the subroutines are limited to 8 pages in length. This
means that this section of the main program must end AFTER &h82DA + &h7F8 =(&h204
0=8pages). I.E. &h8AD2
1080 REM To run this example you will need to do the following....
1090 REM SAVE this program to disk.
1100 REM Use GOTO 2 to see where the subroutines must be loaded to and also see
where the end of this program is. WRITE IT DOWN ''
1110 REM Then delete lines 1000 to the end.
1120 REM Now add in subroutine #1, starting at line 1000 (Don't forget the RETURN
'')
1130 REM Use GOTO 2 again to get the start and finish of the new subroutine (and
check that it's not longer than the original in the main program.)
1140 REM Now BSAVE"sub1",&H[start],&H[finish]
1150 REM Repeat the last three steps (from line 1110) for sub2, sub3, sub4
1160 REM Now load in the ORIGINAL program (This one) and run it.
1170 REM If you try to run the program with sub4 still in it and any of the othe
r subroutines saved are longer than sub4 IT WILL PROBABLY CRASH ''
1180 REM All these REM statements have extended this original subroutine to over
8.5 pages so your new subroutines can be upto 8 pages long. Note that this is t

```

he length IN MEMORY, not in bytes shown on-screen.
2000 RETURN

1000 PRINT "This is subroutine # 1
1010 PRINT "Note that the previously defined strings, ¹⁰⁰NOMS and NOMS can still
1 be used "INVS" like this "NOMS" as can any other variables
1020 PRINT "Press any key": AS=INPUT\$(1)
1030 RETURN

1000 PRINT " This is subroutine # 2
1010 PRINT INVS "This would be a good way to load a lot of variables or data into
memory and then reclaim the space they use in the main program" NOMS
1020 PRINT
1030 PRINT "Press any key": AS=INPUT\$(1)
1040 RETURN

1000 PRINT:PRINT:PRINT
1010 PRINT "You could use this method to load a machine code routine into memo
ry and get rid of the loading routine once it was done
1020 PRINT:PRINT "It would also be useful for those people who like to make
their programs harder to copy
1030 PRINT:PRINT "If the program was split between a lot of files (shown as M/C w
ith the FILES command) and just one small basic program it would discourag
e all but the dedicated!!
1040 PRINT INVS "Press any key" NOMS: AS=INPUT\$(1)
1050 RETURN

1000 PRINT "This is subroutine #4
1010 PRINT "If you now list the program you will see that it has been overlaid
and the original section from line 1000 onwards is no longer there.
1020 PRINT "Note that when using this technique once the subroutines have been
saved to disk the main portion of the program, (which calls in the subroutines)
1030 PRINT "must NOT be altered! The addition or subtraction of just ONE BYTE w
ill cause it to crash!!"
1040 PRINT "The purpose of the technique is, of course, to save memory. Long subr
outines, which might only be called once, can be saved on disk instead of in the
program

1050 PRINT "Press any key": AS=INPUT\$(1):CLS
1060 PRINT "In theory, your super-wonderfull new program could consist of a lon
g series of BLOADs and GOSUBs and might not be very long in itself, however you
should have regard for wear & tear on the drive!
070 RETURN